

Лабораторная работа № 1

Абронина Алиса Кирилловна

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Раздел 1

Лабораторная работа № 1

Цель работы

Целью данной лабораторной работы является приобретение практических навыков установки операционной системы на виртуальную машину, настройки минимально необходимых для дальнейшей работы сервисов.

1 Установка Rocky

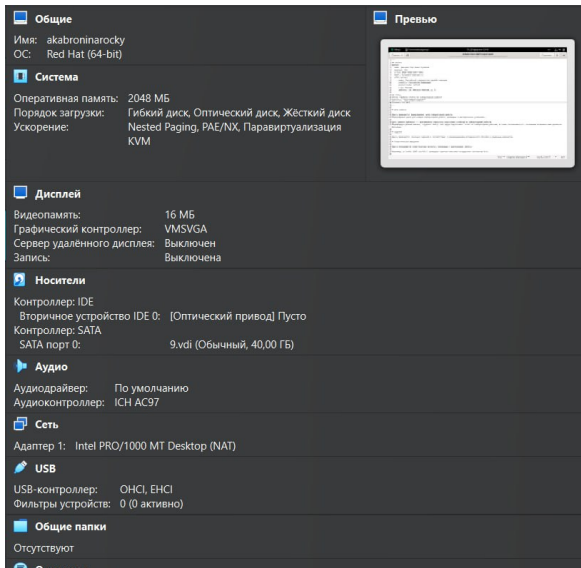
Задание

- 1 Установка Rocky
- 2 Работа с терминалом

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- 3 Контрольные вопросы

Выполнение лабораторной работы

Установка Rocky



Далее устанавливаем имя пользователя, название хоста, языки, страну и др

LOCALIZATION

 **Keyboard**
English (US), Russian

 **Language Support**
English (United States)

 **Time & Date**
Europe/Moscow timezone

USER SETTINGS

 **Root Password**
Root password is set

 **User Creation**
User akabronina will be created

SOFTWARE

 **Installation Source**
Local media

 **Software Selection**
Server with GUI

SYSTEM

 **Installation Destination**
Automatic partitioning selected

 **KDUMP**
Kdump is disabled

 **Network & Host Name**
Connected: enp0s3

 **Security Profile**
No profile selected

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Работа с терминалом

Команда dmesg

```
akabronina@user:~  
akabronina@user ~]$ dmesg  
0.000000] Linux version 5.14.0-611.5.1.el9_7.x86_64 (mockbuild@iad1-prod-build001.bld.equ.rockylinux.org) (gcc (GCC) 11.5  
0 20240719 (Red Hat 11.5.0-11), GNU ld version 2.35.2-67.el9) #1 SMP PREEMPT_DYNAMIC Tue Nov 11 22:20:27 UTC 2025  
0.000000] The list of certified hardware and cloud instances for Enterprise Linux 9 can be viewed at the Red Hat Ecosyste  
Catalog, https://catalog.redhat.com.  
0.000000] Command line: BOOT_IMAGE=(hd0,msdos1)/vmlinuz-5.14.0-611.5.1.el9_7.x86_64 root=/dev/mapper/rl_vbox-root ro resu  
e=/dev/mapper/rl_vbox-swap rd.lvm.lv=rl_vbox/root rd.lvm.lv=rl_vbox/swap rhgb quiet  
0.000000] [Firmware Bug]: TSC doesn't count with P0 frequency!  
0.000000] BIOS-provided physical RAM map:  
0.000000] BIOS-e820: [mem 0x0000000000000000-0x0000000000009fbff] usable  
0.000000] BIOS-e820: [mem 0x0000000000009fc00-0x0000000000009ffff] reserved  
0.000000] BIOS-e820: [mem 0x000000000000f0000-0x000000000000ffffff] reserved  
0.000000] BIOS-e820: [mem 0x00000000000100000-0x000000000007ffeffff] usable  
0.000000] BIOS-e820: [mem 0x000000000007fff0000-0x000000000007fffffff] ACPI data  
0.000000] BIOS-e820: [mem 0x00000000fec000000-0x00000000fec00ffff] reserved  
0.000000] BIOS-e820: [mem 0x00000000fee000000-0x00000000fee00ffff] reserved  
0.000000] BIOS-e820: [mem 0x00000000fffc00000-0x00000000fffffffffff] reserved  
0.000000] NX (Execute Disable) protection: active  
0.000000] APIC: Static calls initialized  
0.000000] SMBIOS 2.5 present.  
0.000000] DMI: innotek GmbH VirtualBox/VirtualBox, BIOS VirtualBox 12/01/2006  
0.000000] Hypervisor detected: KVM  
0.000000] kvm-clock: Using msrs 4b564d01 and 4b564d00  
0.000006] kvm-clock: using sched offset of 10998307929 cycles  
0.000012] clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x1cd42e4dffb, max_idle_ns: 881590591483 ns  
0.000019] tsc: Detected 1899.955 MHz processor  
0.002257] e820: update [mem 0x00000000-0x00000ffff] usable ==> reserved
```

akabronina@user:~

```
[ 43.103394] rfkill: input handler disabled
[akabronina@user ~]$ dmesg | grep -i "Linux version"
[ 0.000000] Linux version 5.14.0-611.5.1.el9_7.x86_64 (mockbuild@iad1-prod-build001.bld.equ.rockylinux.org) (gcc (GCC) 11.5
.0 20240719 (Red Hat 11.5.0-11), GNU ld version 2.35.2-67.el9) #1 SMP PREEMPT_DYNAMIC Tue Nov 11 22:20:27 UTC 2025
[akabronina@user ~]$ dmesg | grep -i "Detected.*MHz"
[ 0.000019] tsc: Detected 1899.955 MHz processor
[akabronina@user ~]$ dmesg | grep -i "CPU0"
[ 0.210015] smpboot: CPU0: AMD Ryzen 7 5800U with Radeon Graphics (family: 0x19, model: 0x50, stepping: 0x0)
[akabronina@user ~]$ dmesg | grep -i "Memory"
[ 0.002606] ACPI: Reserving FACP table memory at [mem 0x7fff00f0-0x7fff01e3]
[ 0.002607] ACPI: Reserving DSDT table memory at [mem 0x7fff0610-0x7fff2962]
[ 0.002608] ACPI: Reserving FACS table memory at [mem 0x7fff0200-0x7fff023f]
[ 0.002609] ACPI: Reserving FACS table memory at [mem 0x7fff0200-0x7fff023f]
[ 0.002609] ACPI: Reserving APIC table memory at [mem 0x7fff0240-0x7fff0293]
[ 0.002610] ACPI: Reserving SSDT table memory at [mem 0x7fff02a0-0x7fff060b]
[ 0.003588] Early memory node ranges
[ 0.007719] PM: hibernation: Registered nosave memory: [mem 0x00000000-0x00000fff]
[ 0.007720] PM: hibernation: Registered nosave memory: [mem 0x0009f000-0x0009ffff]
[ 0.007721] PM: hibernation: Registered nosave memory: [mem 0x000a0000-0x000effff]
[ 0.007721] PM: hibernation: Registered nosave memory: [mem 0x000f0000-0x000fffff]
[ 0.007722] PM: hibernation: Registered nosave memory: [mem 0x7fff0000-0x7fffffff]
[ 0.106612] Freeing SMP alternatives memory: 40K
[ 0.212612] Memory: 1939816K/2096696K available (16384K kernel code, 5783K rwdara, 13988K rodata, 4068K init, 7312K bss, 15
3668K reserved, 0K cma-reserved)
[ 0.212612] x86/mm: Memory block size: 128MB
[ 1.454724] Freeing initrd memory: 61084K
[ 1.486377] Non-volatile memory driver v1.3
[ 1.802732] Freeing unused decrypted memory: 2028K
[ 1.804701] Freeing unused kernel image (initmem) memory: 4068K
[ 1.805096] Freeing unused kernel image (rodata/data gap) memory: 348K
[ 4.055004] vmwgfx 0000:00:02.0: [drm] Legacy memory limits: VRAM = 16384 KiB, FIFO = 2048 KiB, surface = 507904 KiB
[ 4.055013] vmwgfx 0000:00:02.0: [drm] Maximum display memory size is 16384 KiB
[akabronina@user ~]$ dmesg | grep -i "hypervisor detected"
[ 0.000000] Hypervisor detected: KVM
```

```
[ 0.007722] PM: hibernation: Registered nosave memory: [mem 0x7fff0000-0x7fffffff]
[ 0.106612] Freeing SMP alternatives memory: 40K
[ 0.212612] Memory: 1939816K/2096696K available (16384K kernel code, 5783K rwd data, 13988K rodata, 4068K init, 7312K bss, 15
3668K reserved, 0K cma-reserved)
[ 0.212612] x86/mm: Memory block size: 128MB
[ 1.454724] Freeing initrd memory: 61084K
[ 1.486377] Non-volatile memory driver v1.3
[ 1.802732] Freeing unused decrypted memory: 2028K
[ 1.804701] Freeing unused kernel image (initmem) memory: 4068K
[ 1.805096] Freeing unused kernel image (rodata/data gap) memory: 348K
[ 4.055004] vmwgfx 0000:00:02.0: [drm] Legacy memory limits: VRAM = 16384 KiB, FIFO = 2048 KiB, surface = 507904 KiB
[ 4.055013] vmwgfx 0000:00:02.0: [drm] Maximum display memory size is 16384 KiB
[akabronina@user ~]$ dmesg | grep -i "hypervisor detected"
[ 0.000000] Hypervisor detected: KVM
[akabronina@user ~]$ dmesg | grep -i "Mounting.*root"
[akabronina@user ~]$ dmesg | grep -i "mount"
[ 0.106612] Mount-cache hash table entries: 4096 (order: 3, 32768 bytes, linear)
[ 0.106612] Mountpoint-cache hash table entries: 4096 (order: 3, 32768 bytes, linear)
[ 6.008275] XFS (dm-0): Mounting V5 Filesystem c3a3f13e-5696-423e-a8e5-b5d9f9cfce97
[ 8.062098] systemd[1]: Set up automount Arbitrary Executable File Formats File System Automount Point.
[ 8.085191] systemd[1]: Mounting Huge Pages File System...
[ 8.100168] systemd[1]: Mounting POSIX Message Queue File System...
[ 8.115080] systemd[1]: Mounting Kernel Debug File System...
[ 8.120819] systemd[1]: Mounting Kernel Trace File System...
[ 8.318005] systemd[1]: Starting Remount Root and Kernel File Systems...
[ 8.458924] systemd[1]: Mounted Huge Pages File System.
[ 8.473439] systemd[1]: Mounted POSIX Message Queue File System.
[ 8.475883] systemd[1]: Mounted Kernel Debug File System.
[ 8.476191] systemd[1]: Mounted Kernel Trace File System.
[ 11.383227] XFS (sda1): Mounting V5 Filesystem 93626b40-7bf5-469c-869d-6cae979ac589
```

Рисунок 5: выполнение команд

Приобрела практические навыки установки операционной системы на виртуальную машину, настройки минимально необходимых для дальнейшей работы сервисов.